MC 501 Extended Codebook and Reliability Protocol: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Research Proposal and Definitions Practice examples: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: the codebook and the reliability protocol as one document with two headed parts. Every example tweet is paraphrased from a real tweet in the archive; none is quoted, so a search cannot lead back to the account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Part 1: Codebook

Study overview

This codebook measures whether tweets about playing in virtual reality (VR) that refer to learning are more often positive toward VR than tweets that do not (H1), which benefits and obstacles of VR the learning tweets name (RQ1), and whether the balance of benefits to obstacles differs between learning and other tweets, the comparison that separates affordance theory from the cognitive-load account (Mayer et al., 2023) (RQ2). The material is English-language tweets, not retweets, that mention VR together with play, from the Play to Learn in VR archive, collected through the Twitter Academic Research API. The time period is calendar year 2021.

Unit of analysis

One tweet: the text the author wrote in that single post. Quoted tweets, links, images, and replies are not part of it. Every variable below is coded on one tweet. Three fields are looked up, not coded: the tweet ID, the posting date, and the like count. The RQ2 balance (benefit only, obstacle only, both, or neither) is computed in R from variables 3 and 4, so it adds no coding decision.

Variable 1: Learning reference

Element	Learning reference
Conceptual definition and source	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR: learning, teaching, studying, practicing, or being trained. The definition is my own: the discourse studies used a word list (Lampropoulos et al., 2022), but in this archive the same words appear in game titles and idioms.

Element	Learning reference
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person learning, teaching, practicing, studying, or being trained in something, and the activity happens in or through VR. Otherwise mark 0. If the tweet contains a learning word, check what the word means in its sentence before marking.
Level of measurement, and what it allows	Nominal. It allows counts, percentages, and a chi-square test, not a mean, because 0 and 1 are labels.
Coded values	0 = no learning reference. 1 = learning reference.
Edge-case decision rules	1. A learning word that is part of a game or product title (a shooting game called a trainer) or an idiom (learned my lesson) is coded 0 unless another clause describes learning.
	2. Learning to play a VR game, or relearning it on new controllers, is coded 1, because the skill is gained in VR.
Validity sentence	I would doubt this variable if most of the tweets coded 1 turned out to be companies advertising learning apps, because it would then be measuring marketing copy rather than people describing learning.

Variable 2: Tone toward VR

Element	Tone toward VR
Conceptual definition and source	The writer's evaluation of VR or of the VR experience the tweet describes: favorable, unfavorable, or neither. This is sentiment as the discourse studies use it (Lampropoulos et al., 2022; Mohammad & Turney, 2013), read here by a person rather than counted by a lexicon.
Operational definition	Find every word or emoji that evaluates VR or a VR experience, including a game played in VR (fun, love; nauseous, broken). Mark 3 when every evaluation is favorable, 1 when every evaluation is unfavorable, and 2 when the tweet has both or has none. Evaluations of something other than VR do not count.
Level of measurement, and what it allows	Ordinal: three ordered categories with no claim that the steps are equal. It allows a median and a chi-square test, not a mean.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.
Edge-case decision rules	1. Praise for a game played in VR counts as an evaluation of a VR experience. Praise for a game played outside VR does not, so a tweet with only that praise is coded 2.
	2. An advertisement with no evaluative word is coded 2; one that calls the experience amazing is coded 3, because the variable records what the tweet says, not sincerity.

Element	Tone toward VR
Validity sentence	I would doubt this variable if the positive codes were mostly promotional tweets, because advertising is favorable by design and says nothing about how users feel.

Variable 3: Affordance named

Element	Affordance named
Conceptual definition and source	The benefit the tweet attributes to VR: something VR lets a person do, or do better. In affordance theory an affordance is what an environment offers a person to do (Gibson, 1979), and the categories follow the benefits users name most often (Allcoat & von Mühlenen, 2018; Laine & Lee, 2024).
Operational definition	Look for a clause that says what VR lets someone do or makes better, and match it to one category below. If the tweet names benefits in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a benefit, mark 0.
Level of measurement, and what it allows	Nominal. It allows counts, percentages, and a chi-square test, not a mean or a median.
Coded values	0 = no benefit named. 1 = fun or engagement (enjoyment, being absorbed). 2 = hands-on interaction (moving, building, handling objects). 3 = doing it with others. 4 = practicing a skill (getting better at something, in a game or outside it). 5 = other benefit.
Edge-case decision rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking) is coded 0. A feature is not a benefit until the tweet says what it is for. 2. Learning to play a VR game better is coded 4, the same as practicing piano in VR, because the variable records the kind of benefit, not whether it carries outside the game.
Validity sentence	I would doubt this variable if more than a third of the benefits landed in "other benefit," because the categories would then miss what users actually name.

Variable 4: Constraint named

Element	Constraint named
Conceptual definition and source	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged someone from doing something in VR. A constraint is this study's counterpart to an affordance (Gibson, 1979), what the environment blocks, and the categories follow the barriers the reviews report most often (Sunardi et al., 2025; van der Meer et al., 2023).

Element	Constraint named
Operational definition	Look for a clause that names a problem with VR that limited or discouraged use, and match it to one category below. Break a tie between two categories as for variable 3. If no clause names a problem, mark 0.
Level of measurement, and what it allows	Nominal, as for variable 3.
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, eye strain, headache). 2 = hardware cost (cannot afford a headset or computer). 3 = technical performance (lag, crashes, glitches, setup trouble). 4 = other obstacle (space, hygiene, other players' behavior, anything else).
Edge-case decision rules	1. "I can't play, no headset yet" is coded 2 only when price is stated; otherwise it is coded 4. 2. Being thrown out of a match by a glitch is coded 3. A problem the writer only jokes about expecting someday, without saying it has limited anyone yet, is coded 0.
Validity sentence	I would doubt this variable if I only ever coded an obstacle when the tweet's tone was negative, because the variable would then be copying tone rather than recording which obstacle was named.

Decision rules for the whole codebook

These rules settle the hard cases met in 30 immersion tweets drawn at random and kept out of the study sample.

1. Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, mark it out of frame, code nothing, and replace it with the next tweet on the reserve list.

2. If two tweets in the sample carry the same text word for word, code the first and replace the second from the reserve list.

3. A tweet that is only a string of hashtags is coded 0 on learning reference, 2 on tone, 0 on affordance, and 0 on constraint.

4. Code what the tweet says, not who sent it: an advertisement is coded by its words like any other tweet. A sexually explicit tweet is marked out of frame, replaced, and never used as an example.

5. When one clause fits two variables, code it on each. A sentence saying friendly strangers taught the writer to play is a learning reference (1) and an affordance of doing it with others (3).

Examples

Category	Two examples, each paraphrased from a tweet in the archive
Learning reference 1	(a) A new headset owner says friendly strangers in a VR tag game taught them how to play.
	(b) A university says VR training lets students practice clinical skills.
Learning reference 0	(a) A promotion invites readers to learn more about an upcoming VR shooting demo.
	(b) A string of technology hashtags that puts machine learning beside VR.
Tone 3, positive	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A gamer calls a VR title one of the most immersive they have tried.
Tone 2, neutral or mixed	(a) A student asks whether a cabled headset will reach the hallway (no evaluation).
	(b) A player says every VR owner must try one game, but it strains the eyes (both).
Tone 1, negative	(a) A player says the motion controllers always bug out, so they rarely use VR.
	(b) A buyer says the headset costs too much for what it plays.
Affordance 0, none	(a) A player is buying the same horror game again to play it in VR.
	(b) A recap of a console maker's announcement stream lists upcoming games, some in VR.
Affordance 1, fun or engagement	(a) A player calls a VR horror game great and spooky.
	(b) A player is excited to be slicing blocks in a rhythm game again.
Affordance 2, hands-on interaction	(a) A writer wonders whether a VR driving game gives the feel of the road and of acceleration.
	(b) A social VR user says VR gets them standing up instead of sitting for hours.
Affordance 3, doing it with others	(a) A player says they want to spend more time in a social VR app with friends.
	(b) A group that cannot meet in person plans to play a tabletop game together in VR.
Affordance 4, practicing a skill	(a) A VR app promises students practice at interviews and public speaking.
	(b) A player advises a newcomer to practice a VR game in multiplayer first to learn it.
Affordance 5, other benefit	(a) A player tells themselves an all-day rhythm game session counts as exercise.
	(b) A player says VR has been a big help to them through lockdown.

Category	Two examples, each paraphrased from a tweet in the archive
Constraint 0, none	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A player calls a VR demo amazing from start to finish.
Constraint 1, physical discomfort	(a) A new headset owner is lying down after an hour of VR tag, feeling motion sick.
	(b) A player says a cockpit with a lot of roll makes them nauseous in VR.
Constraint 2, hardware cost	(a) A player says they play a game without a headset because they cannot afford one.
	(b) A social VR user wants a better headset someday but says they are expensive.
Constraint 3, technical performance	(a) A player says the motion controllers always bug out.
	(b) A player says their boundary setup resets almost every time they start a VR game.
Constraint 4, other obstacle	(a) A student's study room has no space for a cabled headset.
	(b) A child skipped a shared headset because of germs.

Part 2: Reliability protocol

I am the only coder. Every reliability claim in this protocol is a claim about my own consistency over time, intracoder reliability (stability), and not agreement between two coders.

6. The statistic and its threshold

I will report Krippendorff's alpha for each variable, with the nominal metric for learning reference, affordance, and constraint and the ordinal metric for tone. I chose it over Cohen's kappa because one statistic then covers all four variables and respects tone's order, so a shift from positive to neutral counts as a smaller disagreement than a shift from positive to negative. This is intracoder reliability (stability): the two passes are mine, so alpha measures how consistently I apply the codebook, not whether two coders agree. I will also report percent agreement, but not on its own, because it ignores the agreement that chance alone produces (Lombard et al., 2002). That matters here: most tweets name no benefit and no obstacle, so two passes could agree on 0 most of the time while disagreeing on nearly every tweet that does name one.

A variable passes at alpha of .80 or higher, the level Lombard et al. (2002) call acceptable in most situations. A stability figure is a weaker claim than agreement between two coders, because it cannot catch a reading I make the same way both times, so I will report it as stability and never as if it were intercoder agreement.

7. The stability procedure and a fresh sample

I will re-code 15 tweets, 10 percent of the 150, at least 48 hours after their first coding. Before the second pass, I copy those 15 tweet IDs and texts into a new coding form with no code columns, so my first codes stay hidden until both passes are done. I then compute alpha and percent agreement in R from the two forms joined by tweet ID.

The 15 come from tweets that played no part in building or revising the codebook: the 30 immersion tweets and the 25-tweet pilot in the Sampling Plan are both excluded. After I freeze the codebook, I draw the 15 at random in R, with the seed written in the script, from the 125 sample tweets that are not in the pilot: 8 from the learning-word half and 7 from the other half, so both groups H1 compares are represented.

8. When it fails, and reporting a disappointing number

If a variable falls below .80, I revise that variable only. I read every disagreement for it in my disagreement log (the tweet ID, both codes, and the rule each pass applied) and rewrite or add only the rules where disagreements cluster, without changing any category's conceptual definition. Two guards keep the codebook from bending toward my habits: the revised rules are written and dated before I look at any new tweet, and I test them on a new fresh sample of 15 drawn the same way, never on the 15 that failed, then re-code all 150 tweets on that variable. I allow two revision rounds. A variable still below .80 after the second is reported as unreliable and is not used in any test.

I will report every figure I compute, for every variable and every round, including any that miss .80. For a low figure I will say that my own reading of that variable shifted across two days, name which result rests on it (for tone, that is H1), and describe that result as tentative.

9. Single-coder safeguards

1. Rules before coding: I wrote every category's inclusion and exclusion rule in Part 1 before coding the study sample.

2. Examples from my material: each category carries examples paraphrased from real tweets in the archive, and I check a hard case against them before coding it.

3. A frozen codebook with a change log: I freeze the codebook after the pilot and log every later change with its date, the rule changed, and the reason.

4. A fixed order with the source hidden: I code in tweet-ID order with the username, follower count, and like count hidden until all four variables are coded.

10. Second-coder counterfactual

With a second coder, I would double-code 30 tweets, 20 percent of the sample, drawn the same way as the fresh reliability sample, and report Krippendorff's alpha for each variable

as intercoder agreement. I would train the coder on this codebook with 20 practice tweets from outside the sample, meet to compare codes, and settle each disagreement against the written rule, revising the rule if the rule caused it. The finding most at risk is H1, because it rests on tone, the variable that needs the most judgment, and the line between neutral and mixed is where two coders would most often part.

References

- Allcoat, D., & von Mühlenen, A. (2018). Learning in virtual reality: Effects on performance, emotion and engagement. *Research in Learning Technology, 26*. https://doi.org/10.25304/rlt.v26.2140
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
- Laine, T. H., & Lee, W. (2024). Collaborative virtual reality in higher education: Students' perceptions on presence, challenges, affordances, and potential. *IEEE Transactions on Learning Technologies, 17*, 280-293. https://doi.org/10.1109/TLT.2023.3319628
- Lampropoulos, G., Keramopoulos, E., Diamantaras, K., & Evangelidis, G. (2022). Augmented reality and virtual reality in education: Public perspectives, sentiments, attitudes, and discourses. *Education Sciences, 12*(11), Article 798. https://doi.org/10.3390/educsci12110798
- Lombard, M., Snyder-Duch, J., & Bracken, C. C. (2002). Content analysis in mass communication: Assessment and reporting of intercoder reliability. *Human Communication Research, 28*(4), 587-604. https://doi.org/10.1111/j.1468-2958.2002.tb00826.x
- Mayer, R. E., Makransky, G., & Parong, J. (2023). The promise and pitfalls of learning in immersive virtual reality. *International Journal of Human-Computer Interaction, 39*(11), 2229-2238. https://doi.org/10.1080/10447318.2022.2108563
- Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. *Computational Intelligence, 29*(3), 436-465. https://doi.org/10.1111/j.1467-8640.2012.00460.x
- Sunardi, Meyliana, Hendric, S. W. H. L., & Rosmansyah, Y. (2025). Exploring immersive virtual reality in higher education: Research gap and future direction, a scoping review. *IEEE Access, 13*, 76308-76321. https://doi.org/10.1109/ACCESS.2025.3565385
- van der Meer, N., van der Werf, V., Brinkman, W.-P., & Specht, M. (2023). Virtual reality and collaborative learning: A systematic literature review. *Frontiers in Virtual Reality, 4*, Article 1159905. https://doi.org/10.3389/frvir.2023.1159905

Rubric map

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: study overview	4	The study overview paragraph	It names the question (H1, RQ1, and RQ2, with the rival accounts RQ2 separates), the material (English tweets about playing in VR from a named archive and collection route), and the time period (calendar 2021).
Codebook: unit of analysis	10	The unit of analysis paragraph	One coded item is stated exactly, once: one tweet's own text, with what is not part of it named. Every variable, the looked-up fields, and the computed RQ2 balance describe that same tweet.
Codebook: every variable	10	The four variable tables	Every variable carries all six elements from Definitions Practice: conceptual definition with its source, operational definition, level of measurement and what it allows, coded values, edge-case rules, and a validity sentence specific to that variable.
Codebook: decision rules	8	The five decision rules	Each is written as a rule a coder can apply, and each covers a hard call met in the material: off-topic matches, a reposted promotion, hashtag strings, advertisements and explicit text, and one clause that fits two variables.
Codebook: examples	8	The examples table	Every category of all four variables has two examples, each paraphrased from a real tweet in the archive, none invented.
6. The statistic and its threshold	9	Section 6	It names Krippendorff's alpha with the reason for it over Cohen's kappa, calls it intracoder reliability (stability) in those words, and says why percent agreement alone is not enough, with the reason that bites in this data. It gives a threshold (.80) with its source, and the sentence saying a stability figure is a weaker claim than two-coder agreement and is never reported as one.

Rubric row	Points	Where the example earns it	Why it earns the full row
7. The stability procedure and a fresh sample	8	Section 7	All three conditions: 15 tweets, which is 10 percent of the 150; at least 48 hours later; first codes hidden by a fresh form. The sample is kept apart from both sets used to build and revise the codebook, and its size and the way it is drawn are stated.
8. When it fails, and reporting a disappointing number	8	Section 8	It says in advance which categories are revised (only the failing variable's, where disagreements cluster), how (from the disagreement log, rules only), and two guards against fitting the codebook to habits. It commits to reporting every figure, and says what it will say about a low one.
9. Single-coder safeguards	5	Section 9	Four safeguards, above the minimum of three, each named and described as something done in this study.
10. Second-coder counterfactual	5	Section 10	One paragraph with all four parts: 30 tweets double-coded, Krippendorff's alpha reported, training on 20 practice tweets with disagreements settled against the rule, and H1 named as the finding most at risk, with the reason.

The codebook rows total 40 and the protocol rows 35, graded together as one 75-point assignment.

The codebook revises the Definitions Practice entries where the 30 immersion tweets showed a gap: tone now counts a game played in VR as a VR experience, learning reference gained the rule for relearning a game, and constraint gained the rule for problems the writer only expects.